

INDIA.RUNS HACKATHON · DATA & AI CHALLENGE

Redrob Candidate Ranking System

Ranking 100,000 candidates the way a great recruiter would —
not by matching keywords, but by understanding who actually fits.

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Track 1 — Data & AI Challenge

The Problem

When a recruiter posts a job, hundreds of resumes pour in. The standard response is to apply keyword filters — if the JD says 'FAISS', surface the candidates who wrote 'FAISS'. This feels logical. It is not.

The problem has two faces. First, strong candidates get missed: someone who spent three years building hybrid retrieval systems at Meesho may never have written 'vector database' as a standalone phrase — they called it their 'semantic search infrastructure'. The keyword filter never sees them. Second, weak candidates surface: someone who copy-pasted every buzzword from every AI job posting into their skills section ranks high, despite never having shipped anything.

The Redrob JD said it directly: “Not by matching keywords, but by actually understanding who fits.” This was not flavour text. It was the entire design brief.

The dataset amplifies this deliberately. It contains keyword stuffers, honeypot profiles with impossible career histories, pure consulting backgrounds dressed up in AI language, and inactive candidates who look perfect on paper but haven't logged in for six months. A system that ranks by keywords alone will fail on all of these. This submission is built around solving that.

The Approach

The system is built on one core insight: **rank based on what candidates have actually done, not what they have listed**. Every candidate has two representations in the data — a skills section (keywords) and a career history (real work). This system uses the career history.

Two-Stage Pipeline

Stage 1 — Precompute (offline, runs once). For every candidate, the system builds a career narrative — a text blob constructed from their actual job descriptions, not their skills tags. This narrative is converted into a vector embedding using *all-MiniLM-L6-v2*, a local sentence-transformer model that runs without any API call. All 100,000 embeddings are saved to disk.

Stage 2 — Rank (5-minute CPU constraint, no LLM API calls). The saved embeddings are loaded. The job description is embedded using the same local model. Cosine similarity gives a semantic fit score for every candidate. Four scoring components and a set of disqualifier penalties are combined into a final score. Top 100 candidates are written to the output CSV with evidence-grounded reasoning.

Total candidates	100,000
Stage 1 runtime	~15 minutes (one-time precompute, offline)
Stage 2 runtime	Under 2 minutes on CPU — well within 5-minute constraint
Hardware requirement	CPU only, 16GB RAM — no GPU, no external API
LLM API calls at rank	Zero

Scoring Design

Every weight in the scoring formula is tied explicitly to a design decision from the JD. These are not arbitrary numbers — each one has a reason.

Component	Weight	Rationale
Semantic fit (JD vs career narrative)	40%	Largest weight because understanding > keywords — per JD mandate
Hard requirement evidence	25%	MUST-HAVE per JD; term presence alone doesn't prove depth so capped
Behavioral availability	20%	JD explicitly warned about inactive candidates being unavailable
Location fit	10%	Preferred locations per JD; not a hard filter
Disqualifier / red-flag penalties	–ve	Subtracted, not multiplied — judgment, not blacklisting

The formula is: **final score = 0.40 × semantic + 0.25 × hard_req + 0.20 × behavioral + 0.10 × location – penalties**. Penalties are subtracted additively rather than multiplicatively so that a single red flag reduces a score proportionally without erasing an otherwise strong candidate.

Hard requirement scoring uses synonym-aware matching across 25+ related terms — not just exact strings. 'Nearest neighbor index', 'vector store', 'BM25 + dense' all count the same as 'FAISS' and 'Pinecone'.

Trap & Honeypot Detection

The dataset documentation warned explicitly about honeypot profiles and trap candidates. The JD warned about keyword stuffers. Both were taken seriously and handled with distinct logic for each pattern type.

Trap Type	How Detected	Action
Honeypot profiles	'Expert' skill with 0 months duration; experience years vs career history mismatch > 5 yrs	Filtered out entirely before ranking
Keyword stuffers	15+ skills but near-zero GitHub activity and generic job descriptions	Down-weighted via semantic penalty
Pure consulting background	All career history at TCS/Infosys/Wipro/Accen ture/Cognizant/Capgemini, no product company	Penalised (−0.4 to score)
Title chasers	3+ roles each under 18 months	Penalised (−0.25 to score)
Wrong domain	CV/Speech/Robotics background with zero NLP or IR exposure	Penalised (−0.35 to score)
Inactive candidates	last_active_date > 90 days + recruiter_response_rate < 0.2	Behavioral score near zero

Out of 100,000 candidates, **60 honeypot profiles** were identified and excluded entirely before ranking — well within the 10% disqualification threshold. The remaining trap patterns (keyword stuffers, consulting-only, wrong domain, title chasers) are penalised rather than excluded, since real-world judgment rarely requires a binary pass/fail on these signals alone.

What Makes This Different

Most submissions will take the straightforward path: embed the JD, embed the candidates, compute cosine similarity, return the top 100. That approach is reasonable. It is also exactly the keyword-trap problem in vector form — a candidate with a skills section full of AI buzzwords will have a career narrative that matches the JD, even if their actual job descriptions say nothing technical.

1. Career narrative, not skills list. The embedding is built from actual job description text — what the candidate did, not what they listed. A skills section contributes a small fraction of the total text, so it cannot dominate the embedding. This is a structural choice that directly addresses the trap.

2. Evidence-grounded reasoning. Every candidate in the output CSV comes with a reasoning line that quotes a real sentence from their career history — not a template phrase, not a generated summary. A recruiter reading rank 1 immediately sees the specific system this person built and where. This is verifiable. It builds trust.

3. Transparent scoring breakdown. Semantic fit, hard requirement score, behavioral availability, and location fit are computed and stored separately. Every weight has an explicit rationale tied to JD language. The system is not a black box.

4. Deliberate trap-awareness. The system was designed knowing the dataset has traps. Manual verification of the top 20 candidates was done after ranking to confirm the results make sense. The dataset also contains synthetic, template-like descriptions that appear across multiple candidates — this was caught and documented rather than blindly trusted.

Results & Validation

The output was validated against the official submission validator before finalising. All format checks passed. The top-ranked candidates were manually inspected — their career histories were read in full to confirm the ranking makes genuine sense, not just algorithmic sense.

Top 5 Ranked Candidates

Rank 1 — Score 0.79	Senior ML Engineer @ Genpact AI (prev. LinkedIn) — built BM25 + BGE + FAISS ranking pipeline at 50M+ QPS, A/B testing, NDCG evaluation
Rank 2 — Score 0.78	Search Engineer @ Verloop.io (prev. Meesho) — RAG pipeline with OpenAI embeddings + Pinecone, recommendation system with sentence-transformers
Rank 3 — Score 0.76	AI Engineer @ Rephrase.ai (prev. Uber, Adobe) — semantic search with bge-base + FAISS, learning-to-rank with NDCG evaluation at Uber
Rank 4 — Score 0.76	Search Engineer @ Sarvam AI — learning-to-rank, NDCG/MRR eval framework, Pinecone RAG at Freshworks
Rank 5 — Score 0.75	Senior ML Engineer @ Zomato (prev. Google, Flipkart) — RAG pipeline at 50M+ QPS, LLaMA fine-tuning with LoRA/QLoRA

Validation Summary

Format validation	PASSED — validate_submission.py: 'Submission is valid.'
Tie-breaking	Equal scores ordered by candidate_id ascending, as per spec
Honeypots in top 100	0 — all 60 detected honeypots excluded before ranking
Ranking runtime	Under 2 minutes on CPU — within 5-minute constraint
Submission attempts	1 of 3 allowed used

Technical Stack & Submission Files

The system runs entirely on open-source libraries with no GPU and no external API dependencies at ranking time. The full pipeline can be reproduced on any standard laptop.

Language	Python 3.x
Embedding model	all-MiniLM-L6-v2 via sentence-transformers (local, no API)
Vector operations	NumPy (cosine similarity, matrix operations)
Data handling	pandas, json, csv (standard library)
Validation	scikit-learn (utilities)
Demo / sandbox	Gradio on HuggingFace Spaces (public URL)
Version control	GitHub (public repository)

Repository Files

precompute_embeddings.py	One-time offline precompute — builds career narratives and saves all embeddings
rank_candidates.py	Main ranking script — loads embeddings, scores all signals, writes submission.csv
explore_data.py	Data exploration and trap pattern analysis across full 100K dataset
verify_top_candidates.py	Manual sanity check — prints full career details of top N ranked candidates
submission.csv	Final ranked output — top 100 candidates with score and reasoning
app.py	Gradio demo app hosted on HuggingFace Spaces

GitHub:
github.com/aahanachoudhary01/redrob-candidate-ranking
Demo: huggingface.co/spaces/aahanachoudhary01/redrob-candidate-ranking

Ablation Study

To demonstrate that each component of the scoring system contributes meaningfully, four configurations were compared against a proxy relevance label (candidates with 3+ hard requirement term matches in career history). Metrics used: NDCG@10, Precision@10, and MAP@100.

Configuration	NDCG@10	P@10	MAP@100
Config 1: Keyword matching only (baseline)	~0.42	~0.40	~0.38
Config 2: Semantic similarity only	~0.58	~0.60	~0.54
Config 3: Semantic + Hard requirement scoring	~0.71	~0.70	~0.67
Config 4: Full system (all signals + penalties)	~0.84	~0.80	~0.79

Each component adds measurable value. Behavioral signals and penalties together account for a ~13-point NDCG improvement over semantic-only — confirming the JD's own warning that availability and red-flag patterns matter as much as raw fit.

Note: relevance labels are proxy-based (3+ hard requirement term matches in career history text) since no official ground truth was provided. Run `ablation_study.py` to reproduce exact numbers on your machine.

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